PRACTICAL NO 08

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AIM

To implement and analyze various methods of studying correlation and regression including Karl Pearson’s coefficient and Simple Linear Regression.

1. What is Correlation? How is it different from Regression?

Correlation: Correlation is a statistical measure that shows the strength and direction of relations between two variables. Its value ranges from -1 to +1.

+1 → Perfect positive correlation

-1 → Perfect negative correlation

0 → No correlation

Examples:

Study hours ↑ → Marks ↑ (Positive) Price ↑ → Demand ↓ (Negative)

Regression: Regression is used to predict the value of one variable based on another variable.

1. What is Simple Linear Regression? Explain with example.

Simple Linear Regression is a method used to model the relationship between one independent (X) and one dependent variable (Y). It helps in predicting the value of Y using X.

Example:

If study hours increase, marks also increase, then marks can be predicted using a linear equation Marks = 10 × Hours + 30

1. How can we calculate Pearson Correlation Coefficient in Python?

Pearson correlation coefficient measures the linear relationship between two variables.

In Python, it can be calculated using:

pandas: using .corr() function

numpy: using np.corrcoef() function

1. What is Multicollinearity? Does it affect Simple Linear Regression?

Multicollinearity occurs when two or more independent variables are highly correlated with each a dataset. This can make the model unstable and difficult to interpret.

However, in Simple Linear Regression, there is only one independent variable, so multicollinearity not affect it.

1. Can a dataset have a strong correlation but still produce poor predictions using regression? why.

Yes, a dataset can have strong correlation but still give poor predictions.

This may happen due to reasons such as:

The relationship is not truly linear Presence of outliers Small or insufficient data Missing importar variables Noise or errors in data

Even if correlation is high, regression may fail if the assumptions of the model are not satisfied.

In [1]:
```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

In [2]:
```
# Sample dataset
data = {
    'hours': [1, 2, 3, 4, 5],
    'marks': [40, 50, 60, 70, 80]
}

df = pd.DataFrame(data)

# Pearson Correlation
r = df['hours'].corr(df['marks'])
print("Pearson Correlation:", r)
```

```
Pearson Correlation: 1.0
```

In [3]:
```
# Using numpy
r_np = np.corrcoef(df['hours'], df['marks'])[0,1]
print("Numpy Correlation:", r_np)

# Linear Regression
X = df[['hours']]
Y = df['marks']

model = LinearRegression()
model.fit(X, Y)
```

```
Numpy Correlation: 1.0
```

Out[3]:

▾ LinearRegression

LinearRegression()

In [4]:

```python
# Prediction
pred = model.predict(X)

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
```

```
Slope: 9.999999999999998
Intercept: 30.000000000000007
```

In [5]:

```python
# Plot
plt.scatter(X, Y)
plt.plot(X, pred)
plt.xlabel("Hours")
plt.ylabel("Marks")
plt.title("Regression Line")
plt.show()
```